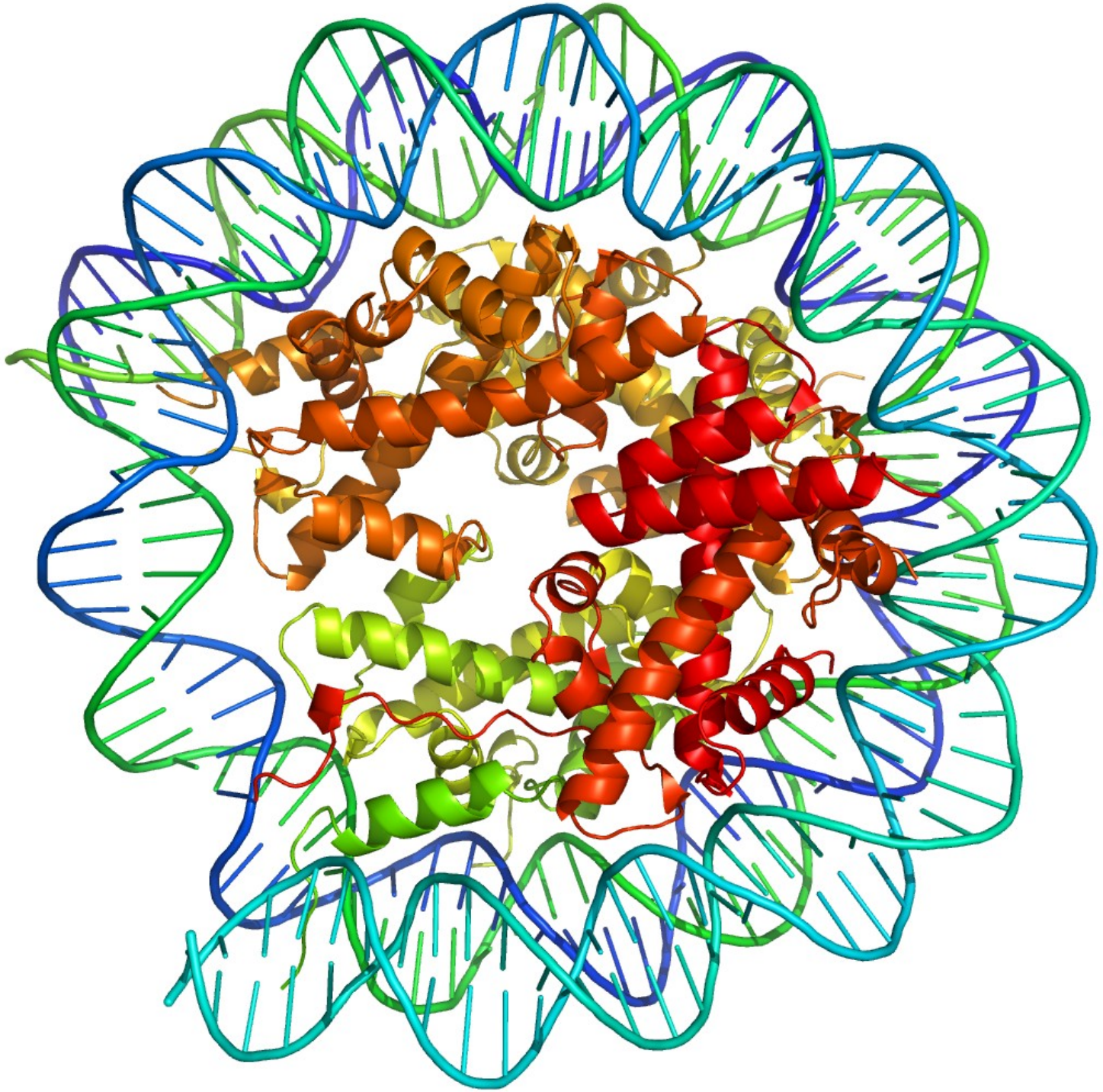
Tunnel Detection in Protein Structures

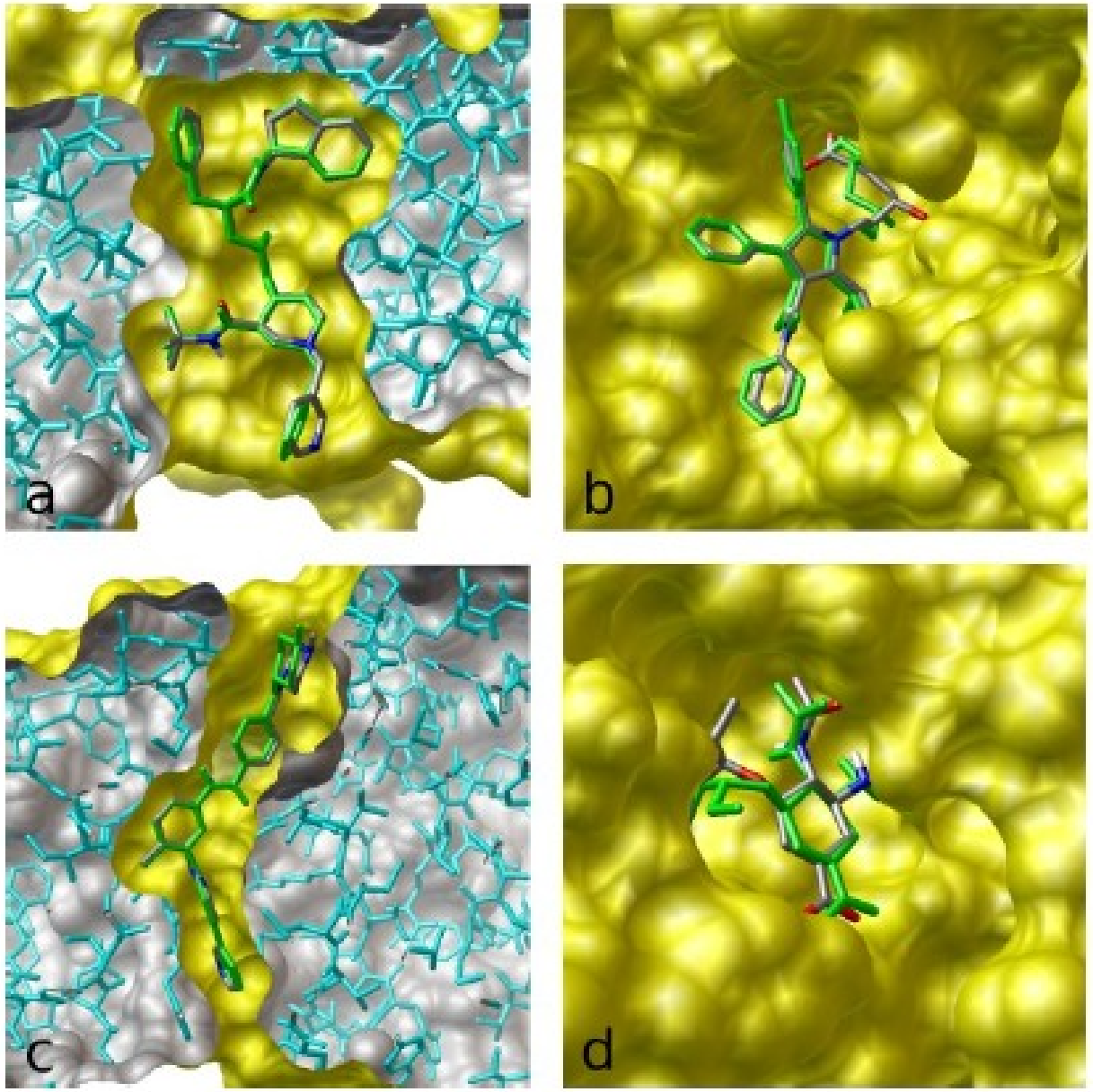
Advanced Topics in Structural Bioinformatics

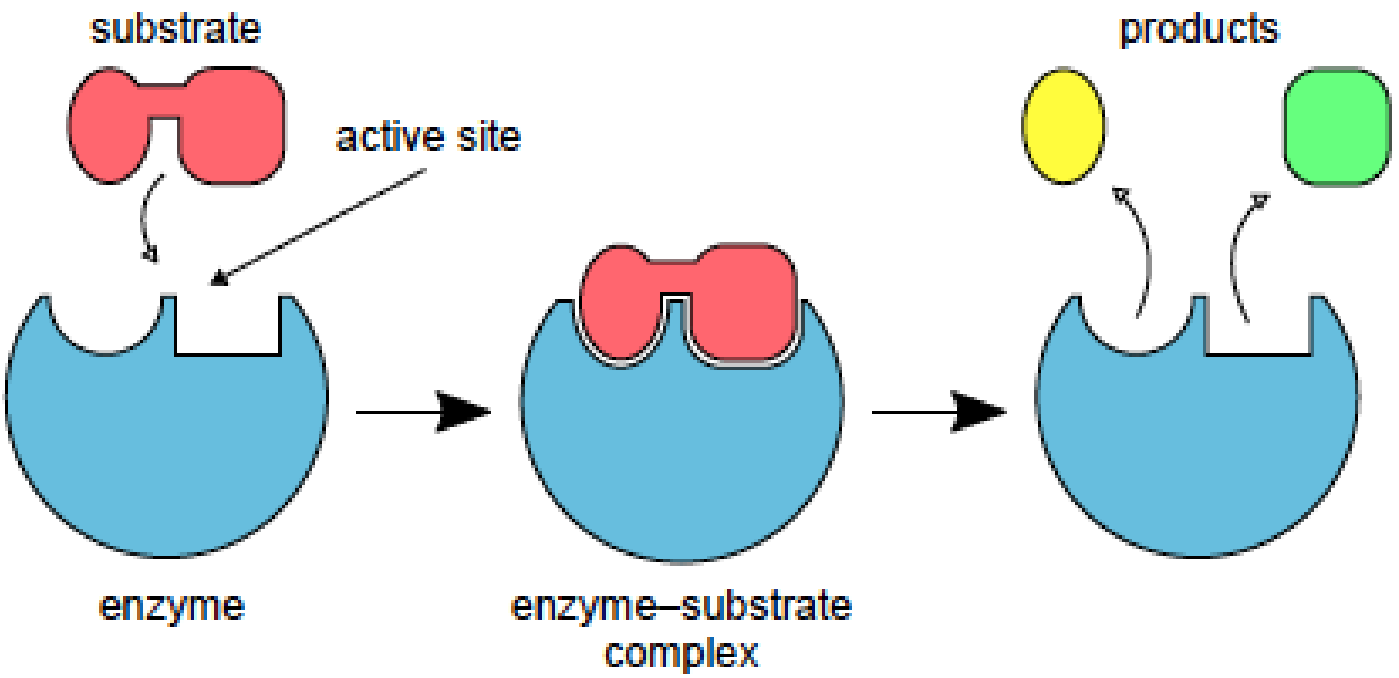
Outline

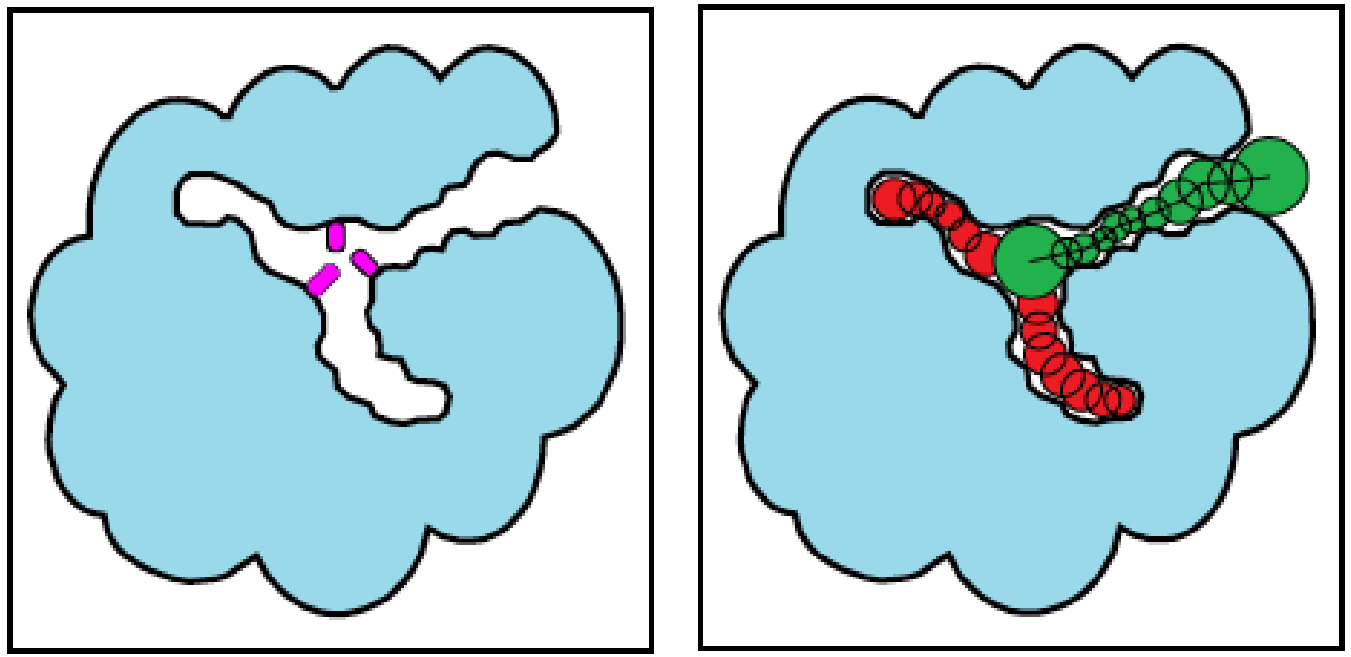
- Problem of motion planning for rigid 3D objects
- Sampling-based motion planning methods like Rapidly Exploring Random Tree (RRT).
- RRT-based method for motion planning of 3D objects.
- Tunnel detection and motion planning in protein structures.
- Tunnel detection in static protein structures.
- Tunnels for spherical probes.
- Tunnel detection in dynamic protein structures.
- Method to analyze traversability of ligands in static tunnels.



The structure of nucleosome complex of a protein (in center) and a DNA fragment (encircling protein), created by software tool PyMOL (PDB le 1AOI).


a
b
c
d

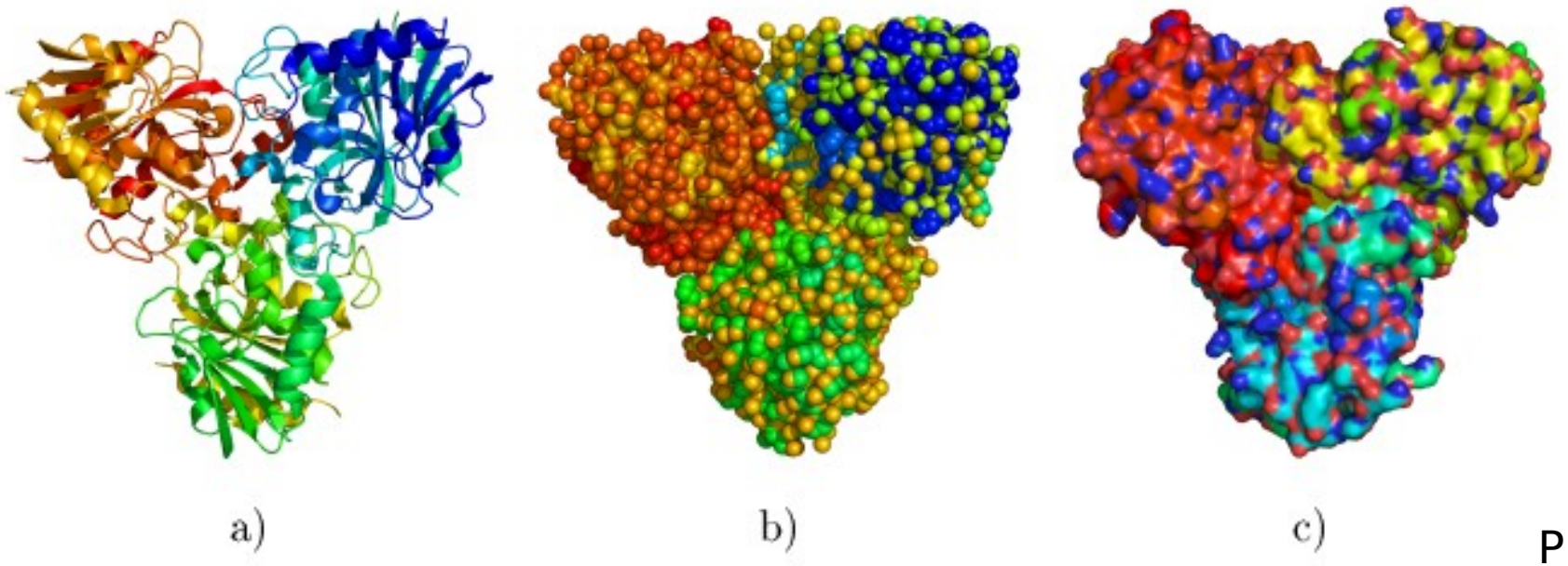

substrate
active site
products
enzyme
enzyme–substrate
complex



a) b)

a) Simplified illustration of the protein (in light blue) with various paths within its structure and highlighted active site, formed by three amino acid residues (in magenta).

b) The result of tunnel detection process, where paths in the structures were scanned with a circular probe. A successfully found tunnel is depicted in green. Paths that were unable to reach the surface of protein are shown in red.



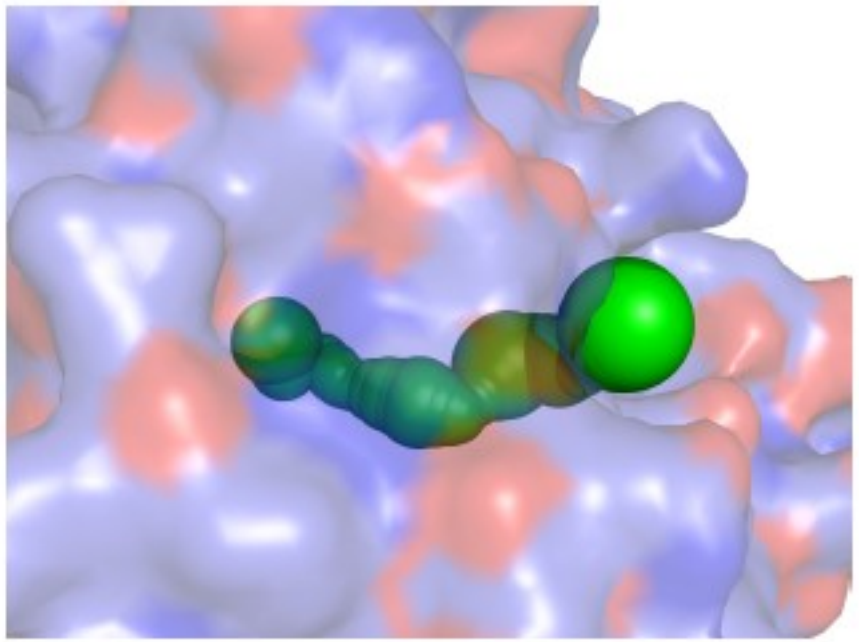
PDB 1LV8

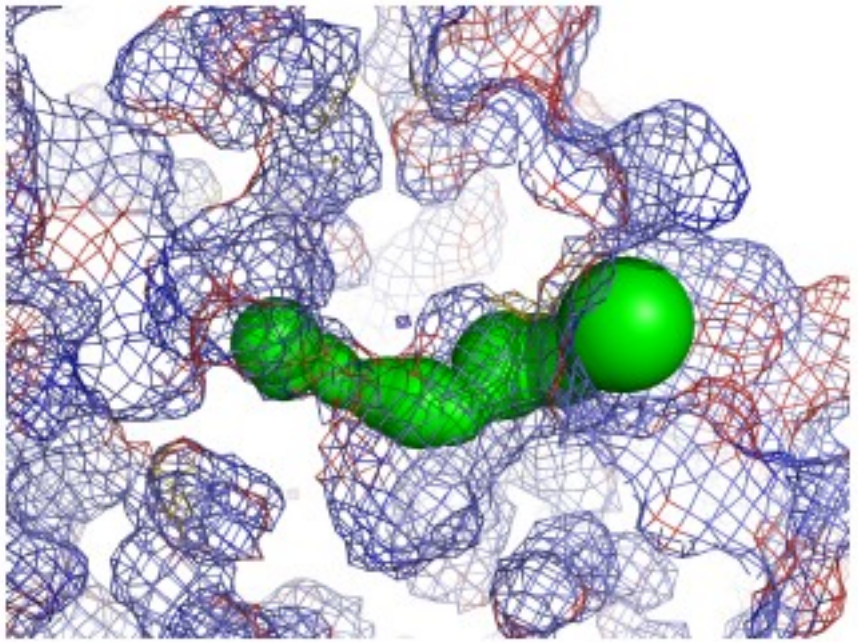
Protein tertiary structure:

a) The cartoon representation shows the secondary structure of protein and provides general protein appearance.

b) The sphere representation depicts all atoms as spheres. This illustration is useful for the tunnel detection task, because spheres are the basis of state space used
in 3D motion planning.

c) The surface representation shows protein surface that would be traced out by the molecules of water in contact with the protein.



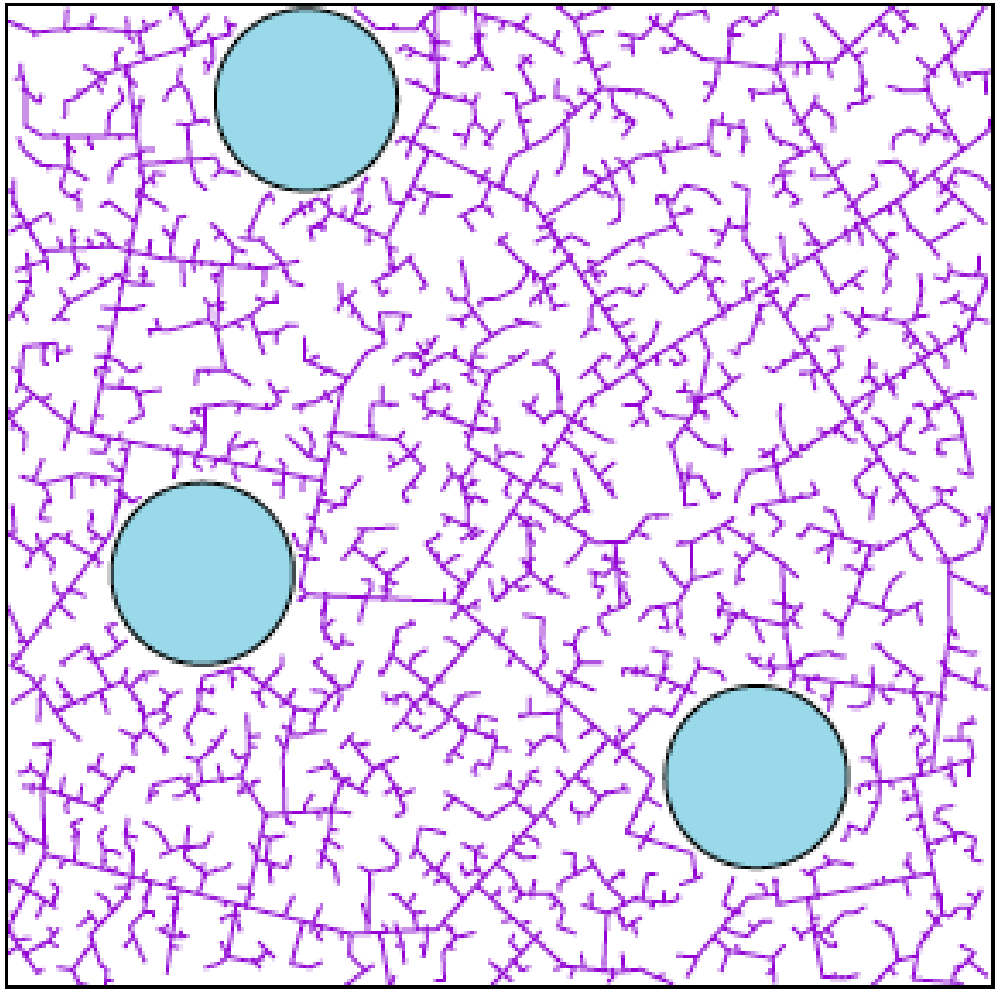


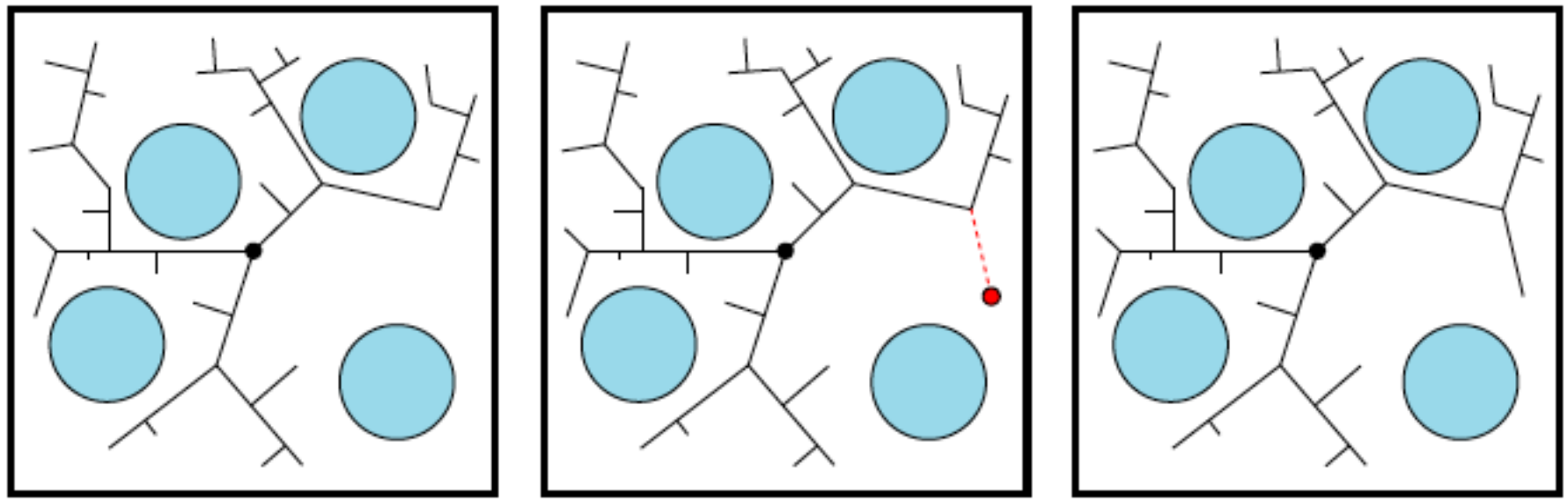
RRT

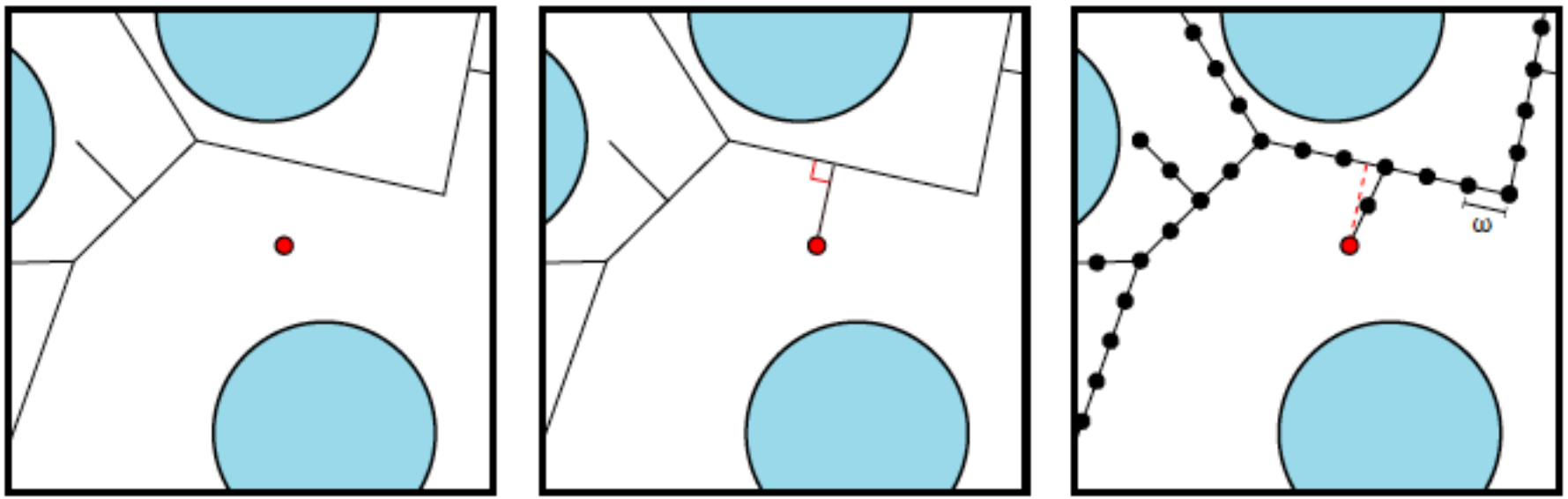
Motion planning
Cybernetics
Robotics
A*
Dijkstra

- Rapidly-exploring Random Tree
- Sampling based algorithm
- Probabilistic
- Randomly building a space filling tree by addition of nodes to collision-free locations
- Can easily handle obstacles
- Great advantage when compared with the complete geometric methods computational demands in high-dimensional assignments

RRT in 2D



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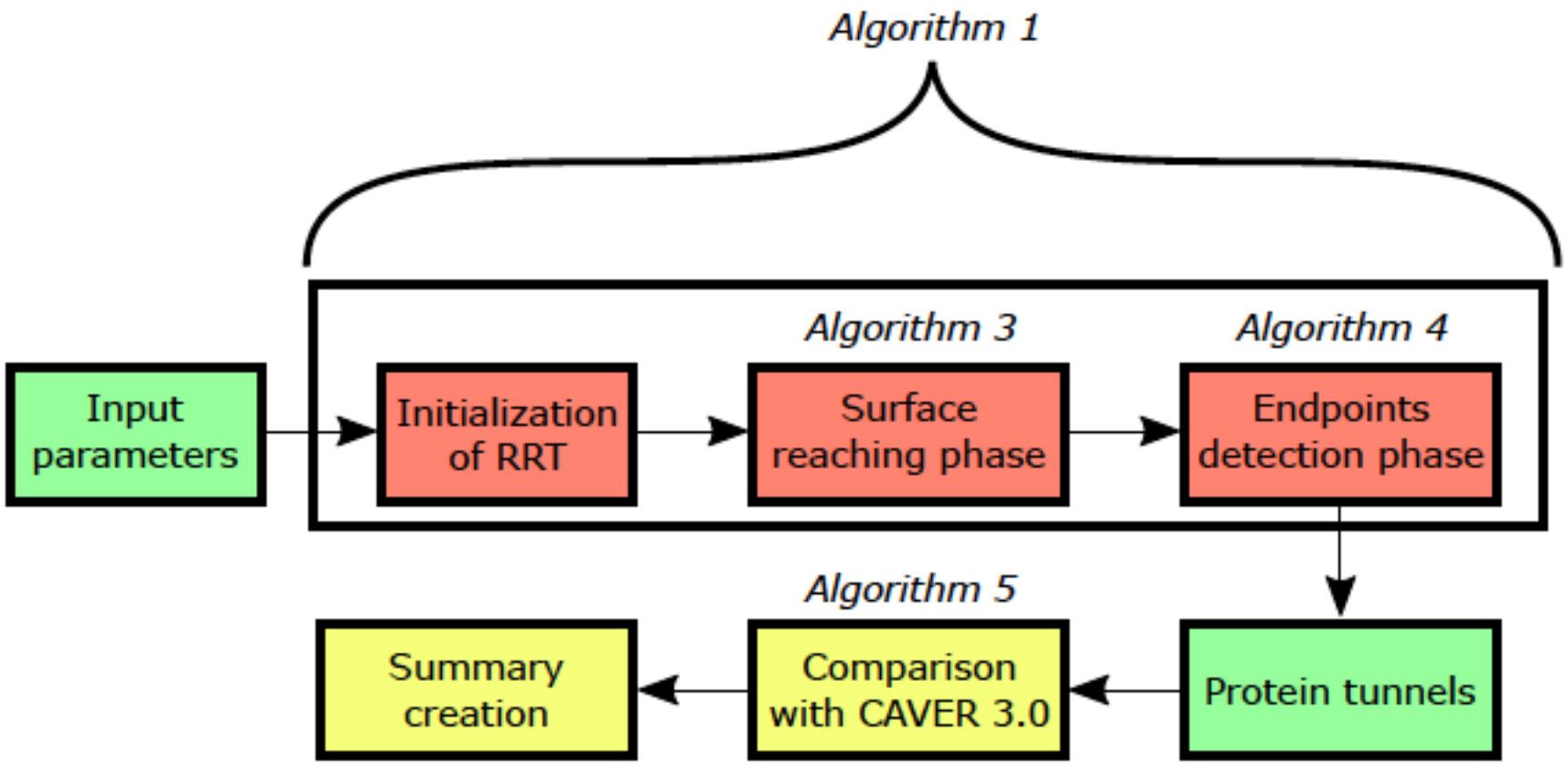
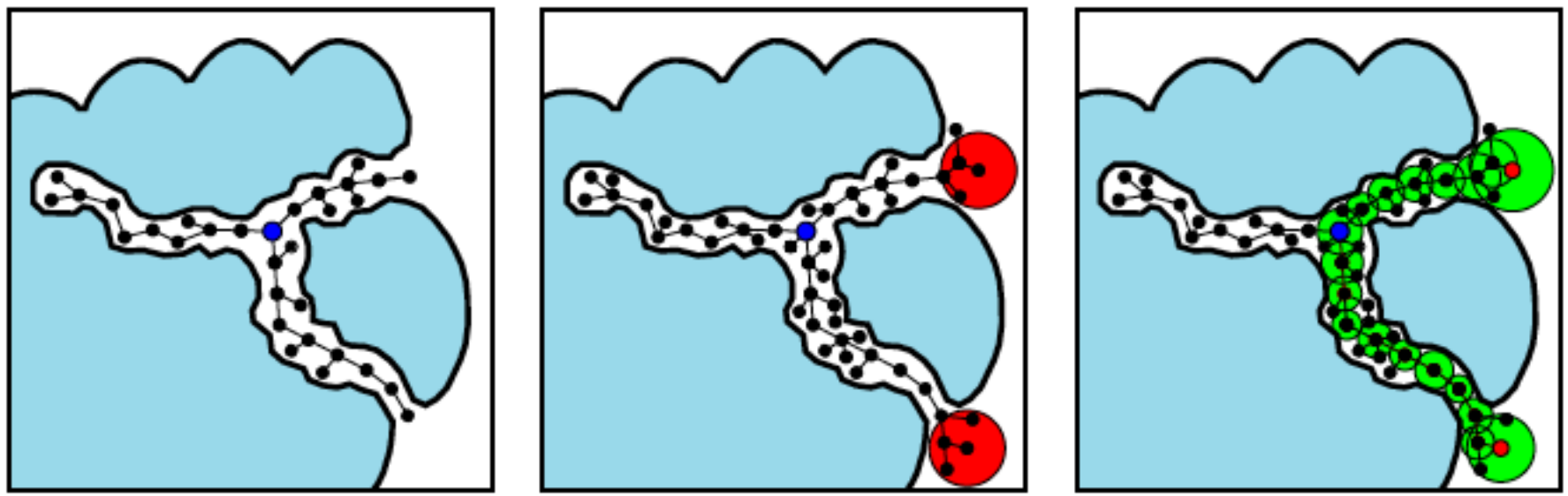
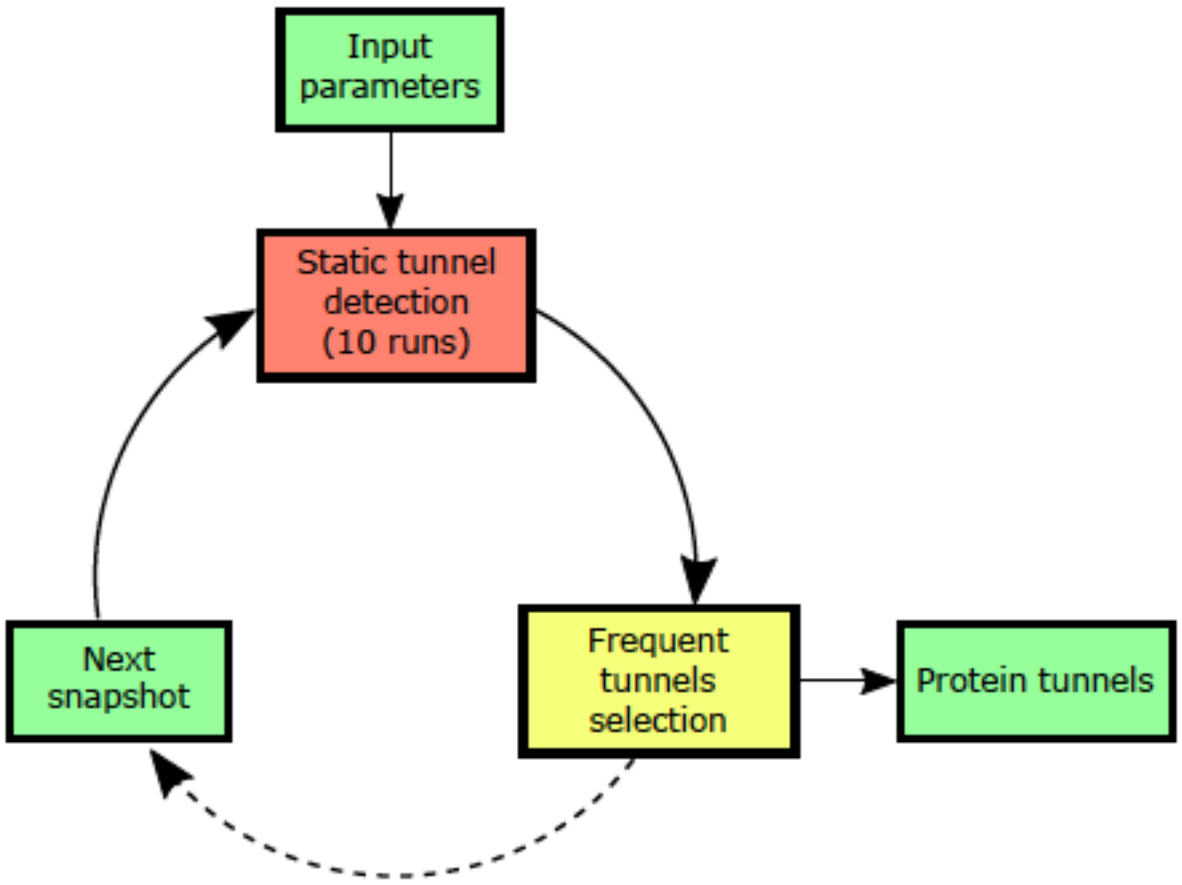

Algorithm 1
Algorithm 3
Algorithm 4
Input parameters
Initialization of RRT
Surface reaching phase
Endpoints detection phase
Algorithm 5
Summary creation
Comparison with CAVER 3.0
Protein tunnels




Input parameters
Static tunnel detection (10 runs)
Frequent tunnels selection
Protein tunnels
Next snapshot

Motion planning and other robotics concepts

- https://github.com/AtsushiSakai/PythonRobotics#k-means-object-clustering

Programming assignment

- Design a meaningful tunnels layout 3 “escape routes” is minimum (1 pt)
- Define an active site (start site of your exploration)
- Adapt the codebase and implement e.g. 20 runs per simulation (2 pts)
- Develop an algorithm to check if all tunnels were detected (2 pts)

https://github.com/RuslanAgishev/motion_planning/blob/master/python_src/rrts/rrt_test2D.ipynb

- https://colab.research.google.com/drive/1YGdFI899MWMjGk6BtHNNOiOQpTIlhrMc?usp=sharing